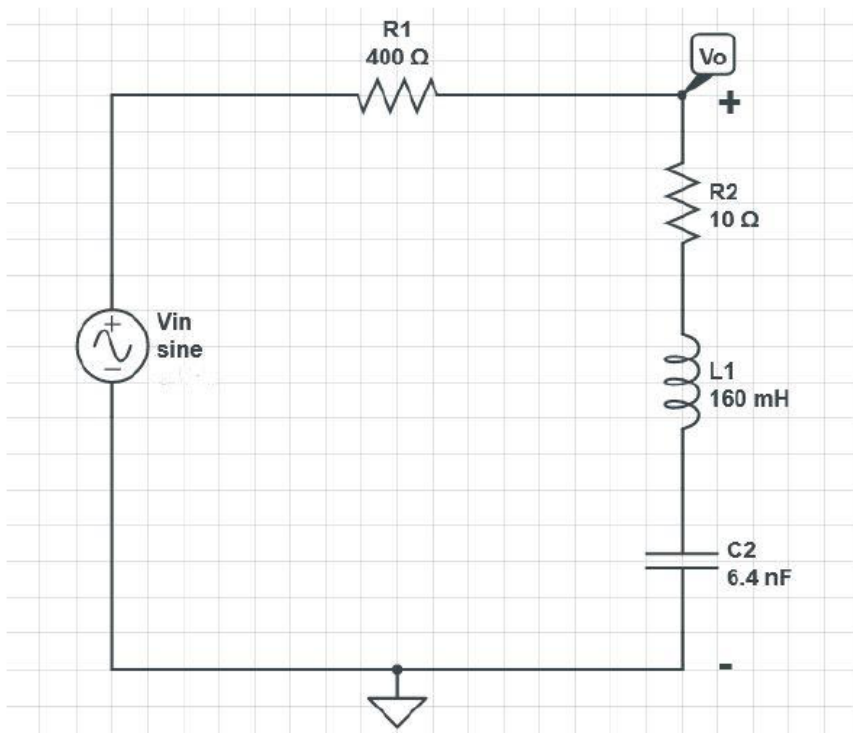


Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Project

Consider a filter having the following circuit:



Here, V_{in} is a sinusoidal ac voltage source having a voltage of $10\text{ V} \angle 0^\circ$. Now, do the tasks given below:

1. Define a filter. **[Marks: 1]**
2. Identify, with reasons, whether the filter mentioned above is passive or active. **[Marks: 1]**
3. Simulate the abovementioned filter in PSpice. Now, get the followings done:
 - a. Construct the circuit in PSpice. **[Marks: 1]**
 - b. Mention the values of the simulation parameters. **[Marks: 1]**
 - c. Illustrate the $|V_o|$ versus f plot. **[Marks: 2]**

d. Also illustrate the *Gain (dB)* versus *f* plot (marking cut-off frequencies). **[Marks: 2]**

e. Complete the following data table: **[Marks: 6]**

The maximum value of $ V_o $ (V)	Resonant frequency, f (kHz)	The value of the quantity that identifies the cutoff frequencies	The cutoff frequencies (kHz)	The bandwidth (kHz)	Selectivity

4. Identify, with reasons, whether the aforementioned filter is low-pass, high-pass, band-pass, or band-stop. **[Marks: 2]**

5. Define pass-band and stop-band and determine the corresponding values for this filter. **[Marks: 3]**

6. Explain the circuit operation of this filter in brief. **[Marks: 3]**

Deliverables:

- A **Report** should be submitted.

The Report should include:

1. Cover Page **[Marks: 1]**
2. Answer to the assigned tasks **[Marks: 22]**
3. Discussion **[Marks: 2]**

- A **Presentation** is required.

For the Presentation, the following marking scheme would be followed:

1. Proper explanation of the workings **[Marks: 5]**
2. Organization of slides **[Marks: 3]**
3. Presentation skill **[Marks: 3]**
4. Response to questions **[Marks: 4]**

Study Material:

Refer to the following book:

Title: Introductory Circuit Analysis
Chapter: Decibels, Filters and Bode Points
Author: Robert L. Boylestad
Publisher: Pearson Education Ltd.

Project
Summer 2025

Course Code: **EEE203L**
Course Title: **Electronic Circuits II Laboratory**

Lab Section: **02**

Date of Submission:

05. 09. 2025

Group No: 3

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Question and Answer no. 01

Q. Define a filter

Answer: A filter is an electronic circuit that allows certain frequencies of a signal to pass while blocking or reducing others.

Low-pass filter: lets low frequencies pass, blocks high.

High-pass filter: lets high frequencies, blocks low.

Band-pass filter: lets only a middle range of frequencies pass.

Band-stop filter: blocks a middle range, lets low & high frequencies pass.

Question and Answer no. 02

Q. Identify, with reasons, whether the filter mentioned above is passive or active.

Answer: For an active filter, there must be a DC power supply and for a passive filter, there must be only R, L, C components. For DC supply rails, there should be an operational amplifier and a transistor component. But, in this circuit, there is no operational amplifier and a transistor whereas it has only R, L, and C components. So, it is a passive filter.

Question and Answer no. 03

Q. Simulate the abovementioned filter in PSpice. Now, get the followings done:

a. Construct the circuit in PSpice.

b. Mention the values of the simulation parameters.

c. Illustrate the $|V_o|$ versus f plot.

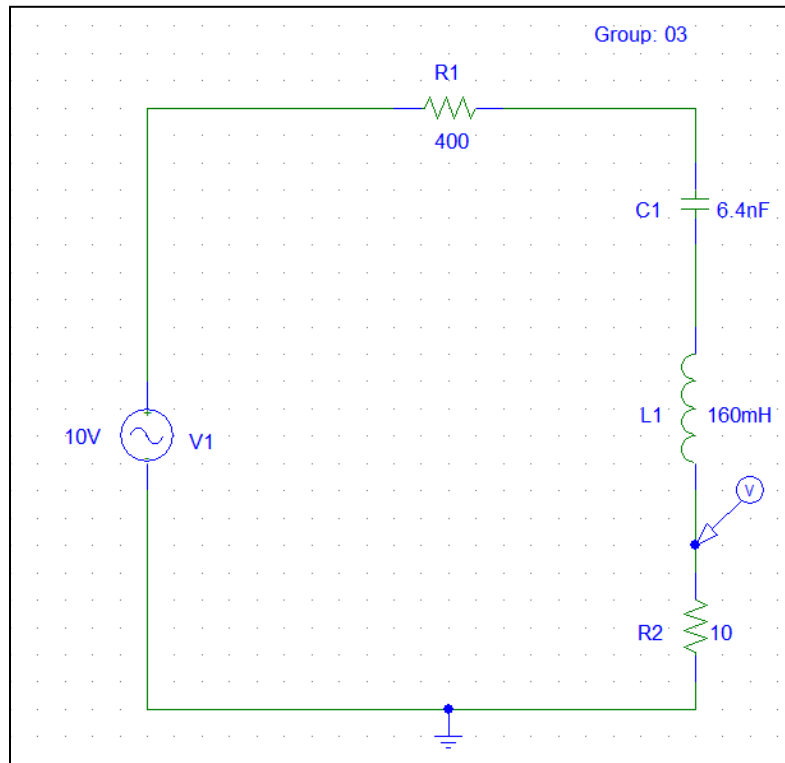
d. Also illustrate the Gain (dB) versus f plot (marking cut-off frequencies).

e. Complete the following data table:

The maximum value of $ V_o $ (V)	Resonant frequency, f (kHz)	The value of the quantity that identifies the cut-off frequency	The cut-off frequency (kHz)	The Bandwidth (kHz)	Selectivity

Answer:

a.



Circuit Diagram

b. V_1 (AC) = $10 \angle 0^\circ$ V;

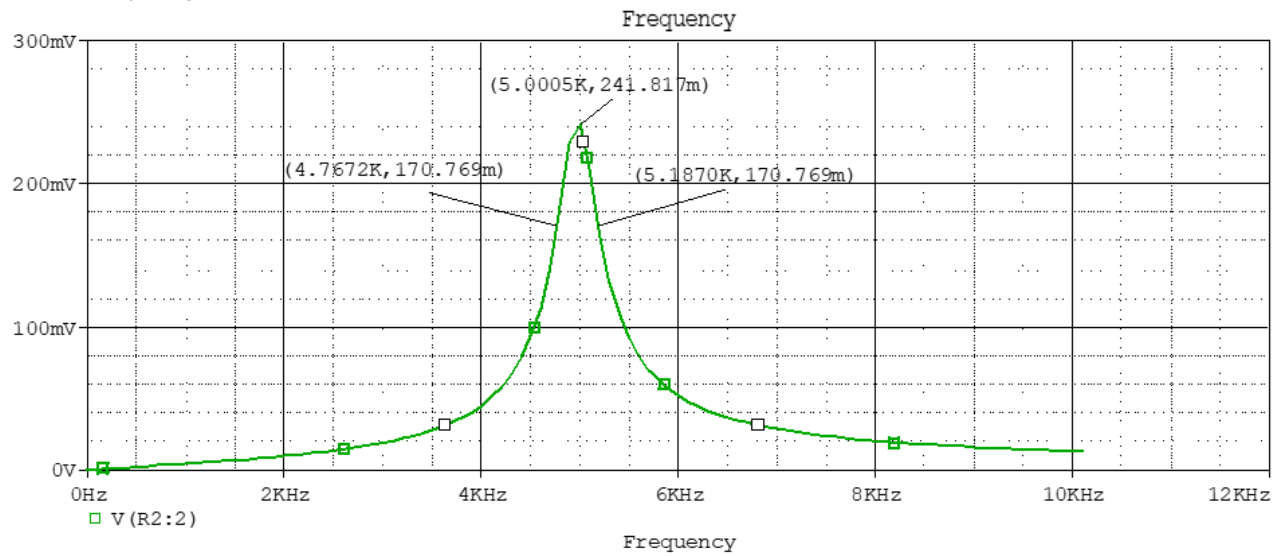
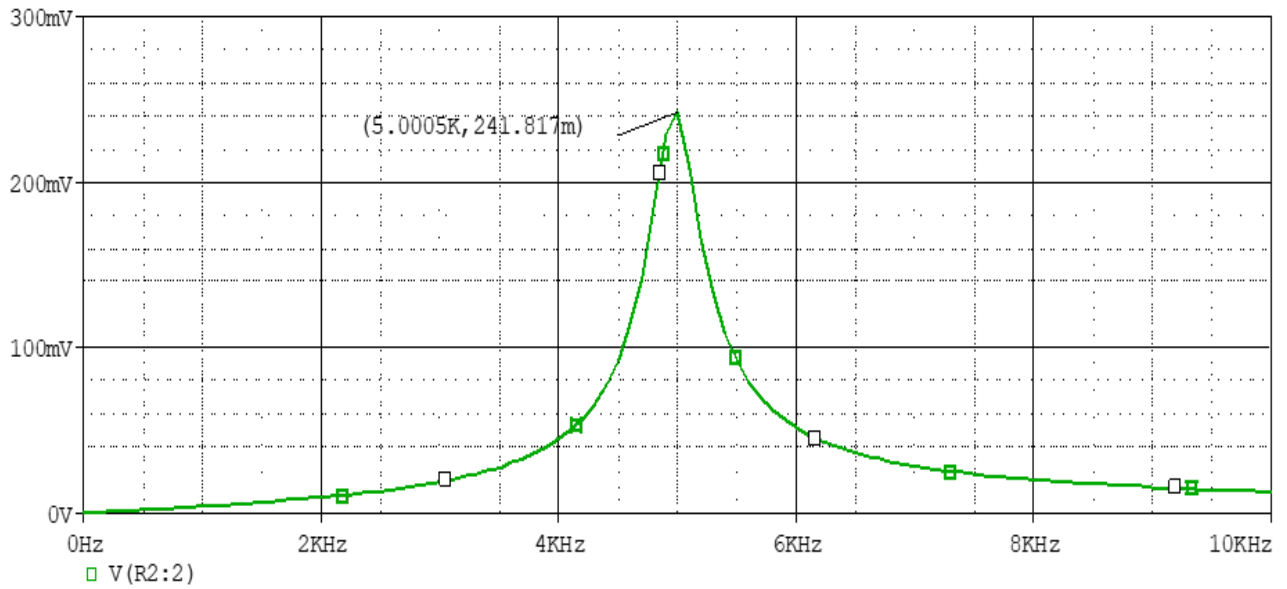
$R_1 = 400 \, \Omega$;

$R_2 = 10 \, \Omega$;

$C = 6.4 \, \text{nF}$;

$L = 160 \, \text{mH}$;

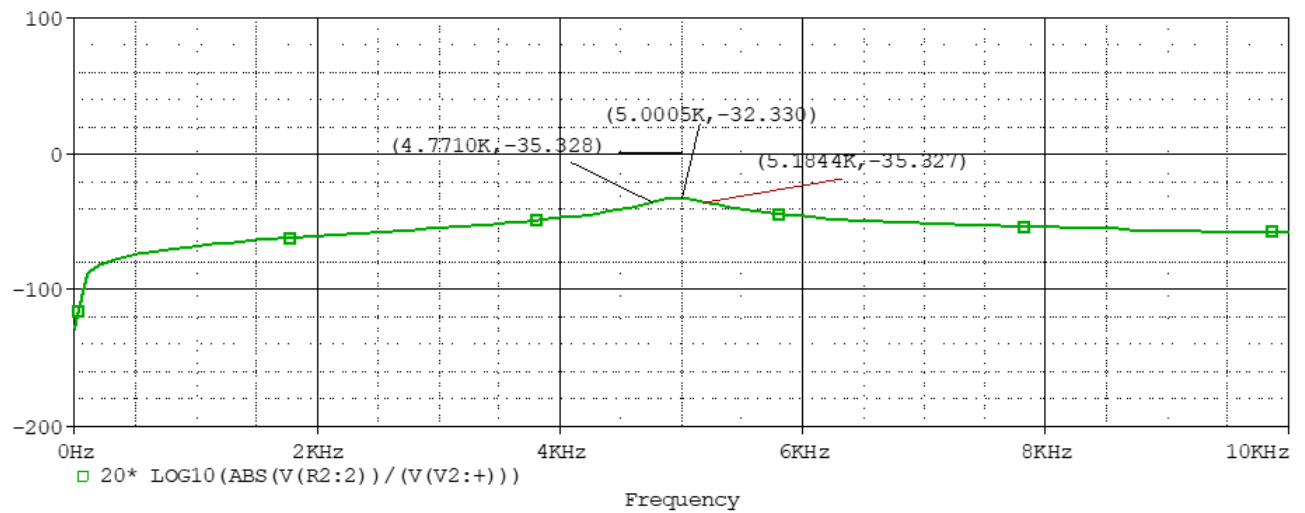
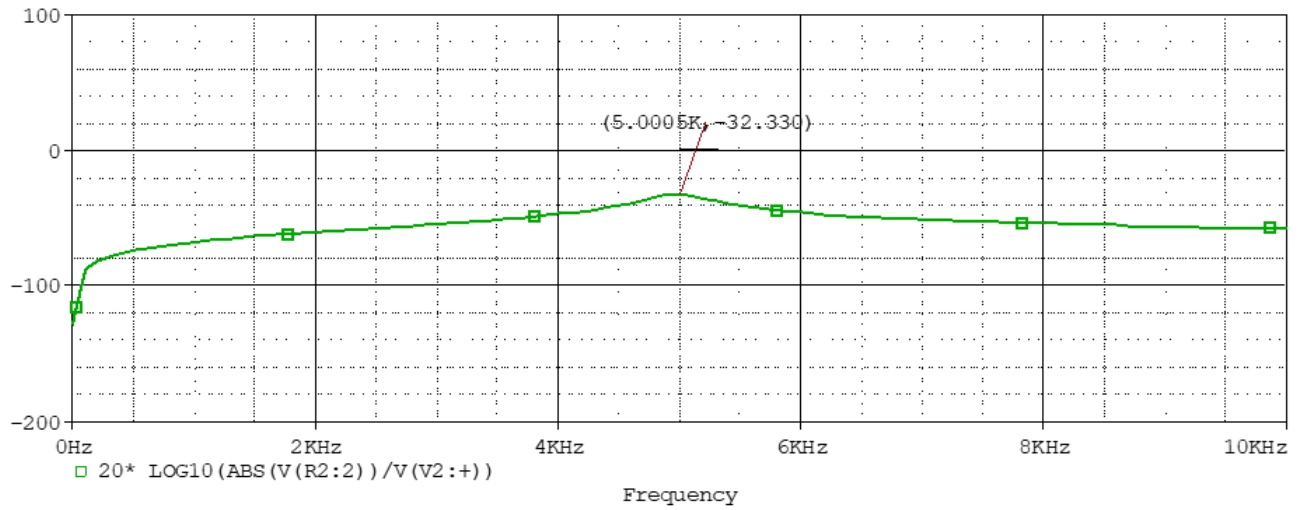
c.



Graph: $|V_o|$ vs f

Here, to find the cut-off frequency, we use the given formula, $|V_o|_{\text{cut-off}} = 0.707 * |V_o|_{\text{max}} = 0.707 * 241.817 = 170.965 \text{ mV}$

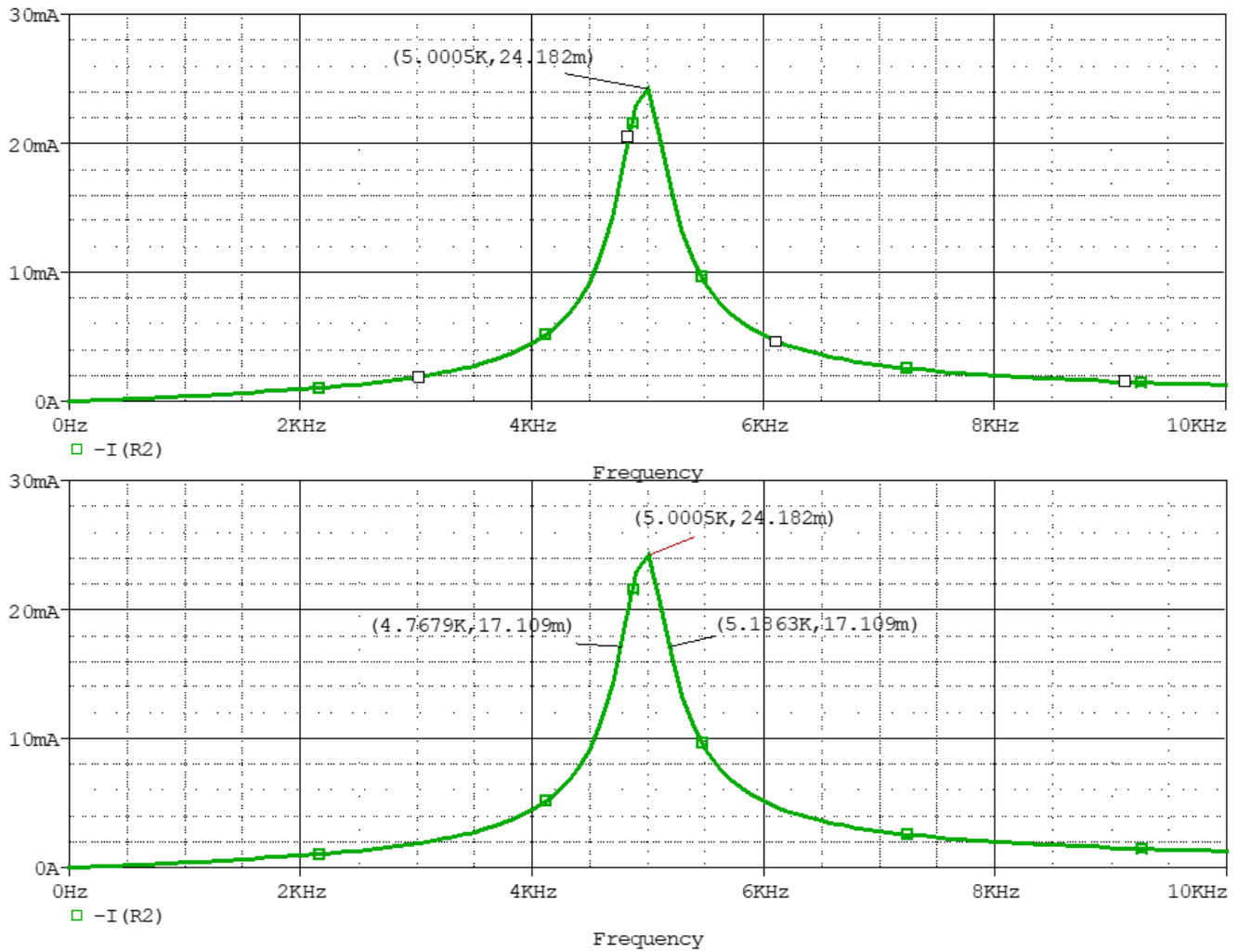
d.



Graph: Gain (db) vs f

Here, to find the cut-off frequency, we use the given formula, $\text{gain(db)}_{\text{cut-off}} = \text{gain(db)} - 3 \text{ db} = (-32.33 - 3) = -35.33 \text{ db}$

e.



Graph: I vs f

The maximum value of $ V_o $ (V)	Resonant frequency, f (kHz)	The value of the quantity that identifies the cut-off frequency	The cut-off frequency (kHz)	The Bandwidth, (kHz) $BW = f_{ch} - f_{cl}$	Selectivity, $Q = f / (f_{ch} - f_{cl})$
0.241817	5.0005	17.096 mA 170.769 mV	$f_{cl} = 4.7679$ $f_{ch} = 5.1863$	0.4184	11.9515

To find the resonant frequency, we plotted the graph of I vs F,

Where $I_{\max}$ is 24.182 mA and corresponding frequency is 5.0005 kHz,

So, the resonant frequency is, $f_r = 5.0005$ kHz

Therefore, to find the cutoff frequency, We calculated $0.707I_{\max} = 0.707 \times 24.182 = 17.096$ mA.

Using the cutoff quantity of 17.096 mA, we identified the closest point in the simulation graph at, 17.109 mA. Which has a low corresponding frequency of $f_{cl} = 4.7679$ kHz

And a high corresponding frequency at, $f_{ch} = 5.1863$ kHz. -

Therefore, The identified bandwidth, $BW = f_{ch} - f_{cl} = 5.1863 - 4.7679 = 0.4184$ kHz.

The Selectivity or Quality factor is defined by, $Q = f_r / BW = 5.0005 / 0.4184 = 11.9515$.

Question and Answer no. 4

Q. Identify, with reasons, whether the aforementioned filter is low-pass, high-pass, band-pass, or band - stop.

Answer: This is a band-pass filter. The given circuit is a series RLC. The output is taken across R_2 . In such circuits, At resonance, the reactances X_L and X_C cancel, current is maximum, so the voltage on R_2 is maximum. At low frequency, the capacitor blocks current (very small output). At high frequency, the inductor blocks current (very small output). This means the output is large only in a certain frequency range (around f_0), which is the hallmark of a band-pass filter.

In $|V_o|$ vs f graph, the curve peaks at approximately 5 KHz. On both sides (lower and higher frequencies), the output falls off. That shape (a resonance peak) is exactly what a band-pass filter looks like.

Low-pass filter: passes only low frequencies which are not seen in this circuit.

High-pass filter: passes only high frequencies which are not seen in this circuit.

Band-stop: blocks a band, passes low and high which are opposite of what we see in this circuit.

Question and Answer no. 05

Q. Define pass-band and stop-band and determine the corresponding values for this filter.

Answer:

Pass-band: frequency range where the output is above the -3 dB level. Here: 4.7678 kHz to 5.1863 kHz.

Stop-band: frequencies where the output is below the -3 dB level. Here: $f < 4.7678$ kHz and $f > 5.1863$ kHz.

Question and Answer no. 06

Q. Explain the circuit operation of this filter in brief.

Answer: In this circuit, it has R, L, and C components and an AC source. So, it is a passive filter as there are no operational amplifiers and transistors so that it can rails to DC supply. The output is taken across R_2 . At low frequency, the capacitor's reactance is high. So, the current and V_o are small in this phase. Similarly, At high frequency, the inductor's reactance is high. So, the current and V_o are small in this phase. At resonance frequency (f_0), X_L and X_C reactive parts cancel. So, the series impedance is minimum ($R_1 + R_2$) and current is maximum. Thus, the voltage on R_2 is maximum. Moving away from f_0 , current drops to the -3 dB points at f_1 and f_2 that sets the bandwidth.

Discussion:

In this project, a series RLC filter was designed and simulated in PSpice to study its frequency response. The circuit consisted only of passive elements (R, L, and C), which confirms that it is a passive filter. The output was taken across a small resistor (R2), making the circuit behave as a band-pass filter. From the simulation results, the output voltage $|V_o|$ was found to be very small at both low and high frequencies. This occurs because, at low frequency, the capacitor blocks current flow, while at high frequency, the inductor blocks current flow. At the resonant frequency (5.0005 kHz), the reactances of the inductor and capacitor cancel each other, leaving only the resistive path. As a result, the circuit current reaches a maximum value, and the voltage across R2 also becomes maximum. The frequency response plot showed a clear peak around resonance, while the gain plot in dB confirmed that the output drops to the -3 dB level at two cutoff frequencies (4.77 kHz and 5.19 kHz). The difference between these frequencies gave a bandwidth of about 0.42 kHz, and the selectivity ratio $Q = f_o / BW$ was found to be around 12, indicating that the filter is fairly selective. The experiment successfully demonstrated the band-pass characteristics of the series RLC circuit. The results from simulation matched well with the theoretical formulas of resonance, cutoff frequencies, and bandwidth. This shows that PSpice can be effectively used to predict and analyze real-world filter behavior before practical implementation.

Reference:

Boylestad, R. L. (2009). *Introductory circuit analysis* (12th ed.). Pearson Education Ltd.